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Clinically relevant predictive modeling for personalized ACL reconstruction classification

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ABSTRACT

Anterior Cruciate Ligament (ACL) reconstruction outcomes and return-to-sport readiness vary significantly among patients, yet current classification methods often lack interpretability and personalization. We propose an explainable predictive model for ACL reconstruction classification through multi-modal analysis of gait dynamics and patient characteristics. Using inertial measurement unit (IMU) sensors on participants' wrists, ankles, and sacrum, we collected gait data during walking and jogging tasks, alongside patient-specific survey information. For gait dynamics, we employed Phase Slope Index to quantify inter-sensor relationships and trained classifiers for different ACL reconstruction outcomes(left vs right injury, healthy vs injured), achieving high classification performance (96.37% accuracy). Model explanations using heatmaps and permutation importance revealed that paired body movements are crucial in classification, with more distinct patterns in jogging than walking. For patient characteristics, t-SNE visualization demonstrated that model confidence correlated strongly with recovery duration. While longer recovery typically leads to more normal gait patterns, our approach provides a quantitative method to visualize this process transparently. This explainable, personalized approach can improve rehabilitation strategies and inform more accurate return-to-sport decisions in sports medicine.

1. Introduction

Anterior Cruciate Ligament (ACL) injuries represent a significant challenge in sports medicine and orthopedics, affecting athletes across various disciplines and skill levels. These injuries not only lead to substantial time away from sport but also increase the risk of early-onset osteoarthritis and other long-term complications (Lohmander et al., 2007). With an estimated 250,000 ACL reconstructions annually in the United States alone, the impact extends beyond individual athletes to team performance, healthcare costs, and long-term quality of life.

Current methods for assessing ACL reconstruction outcomes and determining return-to-sport readiness typically involve clinical examinations, functional tests, and subjective assessments (Abrams et al., 2014). While valuable, these methods often neglect complex movement patterns and lack precision and objectivity, particularly failing to account for the complex interplay between different body segments during movement.

Recent research has focused on quantifying gait measures using IMU sensors and machine learning models. These wearable sensors capture high-frequency three-dimensional motion data, providing precise insights into joint kinematics during functional

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tasks (Picerno, 2017). Many studies derived time-domain features (statistical moments, peak accelerations) and frequency-domain features (power spectral density, wavelet coefficients) to characterize movement patterns (Saboor et al., 2020). Despite promising performance, these advanced analytical methods face limitations in clinical interpretability, with their "black box" nature hindering translation into actionable insights.

To address these limitations, we propose a novel approach integrating multi-modal gait analysis using IMU sensors with patient-specific characteristics. We hypothesized that: (1) paired body segment movements quantified by Phase Slope Index (PSI) matrices would significantly contribute to ACL reconstruction classification; (2) the importance of specific sensor pair combinations would differ between walking and jogging tasks; and (3) patient-specific factors would correlate with classification confidence, revealing insights into temporal dynamics of gait recovery.

2. Related work

This section reviews the current state of research in ACL reconstruction prediction, focusing on IMU-based techniques, multimodal approaches, and the interpretability challenge.

Research has developed predictive models for ACL reconstruction outcomes using IMU data and feature engineering techniques. The engineered features are typically categorized as time-domain features (mean acceleration, RMS values, peak angular velocity) that summarize movement characteristics, and frequency-domain features (derived through Fourier or Wavelet Transforms) that capture periodic components and energy distributions (Duong, 2023). These features represent gait dynamics more accurately, enabling deeper understanding of recovery status and re-injury risks. Leveraging these features, ACL predictive models have demonstrated strong performance in classifying outcomes such as 'successful recovery' versus 'at risk of re-injury' (Saboor et al., 2020).

Building on these single-source approaches, data fusion methods have emerged to optimize ACL predictive model performance by incorporating multiple perspectives beyond IMU data alone. This approach combines information from various sources, such as IMUs with motion capture systems, to provide a more comprehensive view of human movement and knee joint forces (Qiu et al., 2022). Many research efforts have attempted to combine these features using methods ranging from simple concatenation to more advanced techniques such as Kalman filtering or principal component analysis. Nweke et al. (2019). Studies implementing this multimodal approach have reported accuracy improvements of 5% to 15% compared to single-source models (Dehzangi et al., 2017), suggesting that multiple measurement perspectives help identify subtle injury risk indicators.

Despite the strong performance of these models in classifying outcomes, poor interpretability remains a significant barrier to their clinical application. The complexity of machine learning algorithms often creates "black box" models that obscure decision-making processes, challenging clinical trust and implementation (Horst et al., 2019). Efforts to improve interpretability include feature importance scoring to identify which features are most indicative of ACL outcomes, though this approach focuses primarily on global importance, limiting personalized analysis capabilities. For local interpretation, LIME (Local Interpretable Model-agnostic Explanations) has been adopted (Kim et al., 2022), approximating complex models locally with more interpretable models. However, these explanations are valid only for individual predictions and may not generalize well, potentially leading to unstable explanations.

3. Methodology

This section outlines our research methodology addressing poor interpretability in ACL predictive modeling, as shown in Fig. 1. We recruited 79 participants, including 74 ACL patients (31 with left knee injuries, 43 with right knee injuries) and 5 healthy individuals. Data collection utilized five Shimmer IMU sensors placed on participants' bodies (bilateral wrists, ankles, and sacrum), recording accelerometer and gyroscope data at 128 Hz. The protocol consisted of two sequential tasks: walking on a treadmill at 3 mph for 5 min, followed by jogging at 6 mph for 3 min.

For preprocessing, we first synchronized data across sensors and separated walking and jogging sequences through visual inspection of signal pattern changes. We then downsampled the data from 128 Hz to 64 Hz and segmented it into non-overlapping 10-second windows, providing sufficient temporal resolution to capture multiple gait cycles while maintaining computational efficiency. To capture complex interactions between sensor readings, we calculated the Phase Slope Index (PSI) to form a pairwise causality feature matrix. This process involved frequency domain transformation using Fast Fourier Transform, calculating cross-spectral density between sensor dimension pairs, deriving the phase spectrum, computing PSI through linear regression on the phase spectrum across different frequency bands (8 Hz, 16 Hz, 32 Hz), and constructing the pairwise causality matrix. With 5 sensors each providing accelerometer and gyroscope readings across three axes, we had 30 dimensions per window, resulting in 435 unique features representing all possible dimension pairs.

To investigate our first hypothesis and evaluate PSI-based features, we implemented five machine learning models (SVM, Naive Bayes, Random Forest, KNN, and Neural Network) for two binary classification tasks: Left Injured vs. Right Injured and Injured vs. Healthy. We employed 5-fold cross-validation for walking and jogging phases independently, with randomly shuffled data windows. We experimented with three maximum frequency cutoffs (8 Hz, 16 Hz, and 32 Hz) to determine the optimal range for feature extraction.

For model interpretation addressing our second hypothesis regarding task-specific gait adaptations, we calculated permutation importance scores from the trained model, mapped these to their corresponding sensor dimensions, aggregated them by sensor pairs, and visualized them using heatmaps for both walking and jogging conditions. This approach allowed direct comparison of sensor pair importance between tasks, revealing task-specific gait adaptations. To address our third hypothesis concerning patient-specific

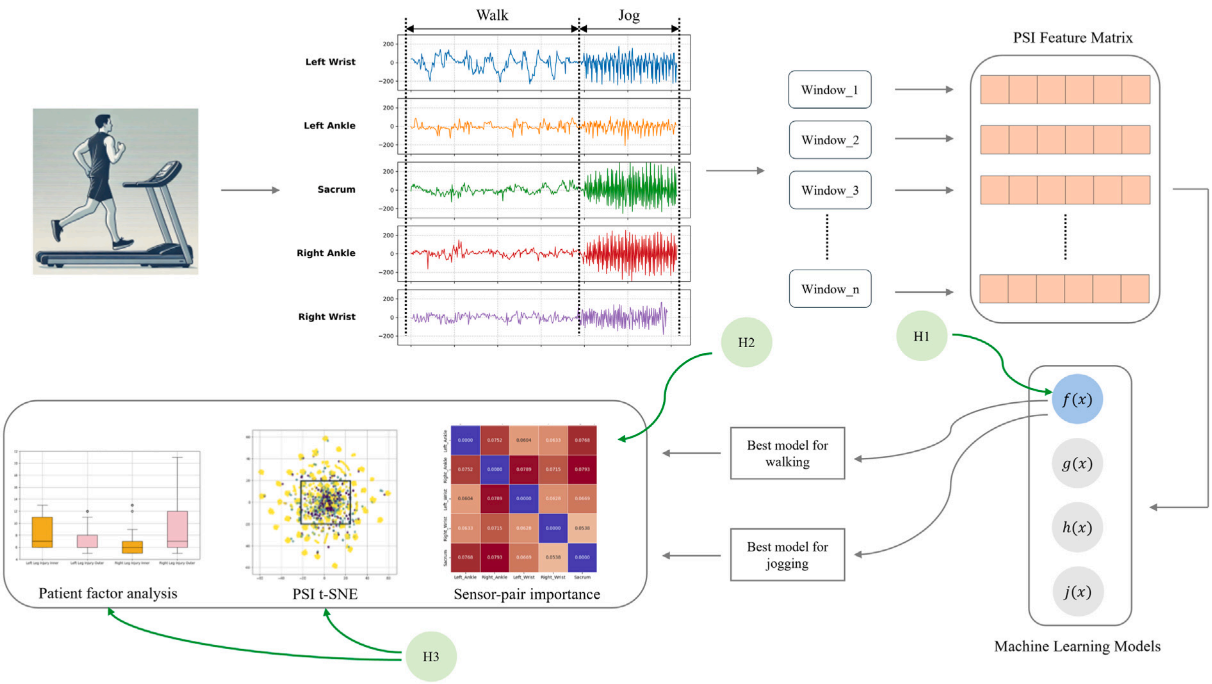

Fig. 1. Five IMU sensors (bilateral wrists, ankles, and sacrum) captured movement during walking and jogging tasks. Data sequences were segmented into 10-second windows with pairwise causality matrices computed using Phase Slope Index (PSI). This feature matrix trained machine learning models for ACL reconstruction outcome prediction. Our hypotheses explored: (H1) whether PSI-based features improve model performance, (H2) if sensor pair importance differs between walking and jogging tasks as shown in the heatmaps, and (H3) whether patient factors correlate with model classification confidence, visualized through t-SNE plots colored by model output probability.

Table 1
Accuracy performance comparison of SVM, Naive Bayes, random forest, KNN, and neural network classifiers on difference ACL reconstruction outcome: (1) left or right injury while walking, (2) left or right injury while jogging, (3) healthy or injured while walking, and (4) healthy or injured while jogging. The results show the KNN model using the 16 Hz low-pass filtered data has the highest overall performance.

Activity phase	Walking						Jogging					
Classification task	Left vs Right			Healthy vs Injured			Left vs Right			Healthy vs Injured		
Frequency Threshold	8 Hz	16 Hz	32 Hz	8 Hz	16 Hz	32 Hz	8 Hz	16 Hz	32 Hz	8 Hz	16 Hz	32 Hz
SVM	0.87	0.85	0.86	0.95	0.95	0.95	0.93	0.96	0.95	0.95	0.95	0.96
Naïve Bayes	0.72	0.75	0.74	0.95	0.94	0.94	0.81	0.79	0.80	0.92	0.95	0.96
Random Forest	0.73	0.71	0.73	0.92	0.92	0.92	0.80	0.78	0.82	0.90	0.90	0.91
KNN	0.93	0.93	0.92	0.98	0.98	0.98	0.96	0.98	0.96	0.96	0.99	0.99
Neural Network	0.86	0.81	0.79	0.95	0.92	0.95	0.85	0.83	0.88	0.95	0.95	0.95

factors and classification confidence correlation, we applied t-SNE for dimensionality reduction of our feature matrix and examined the distribution of classification confidence scores across patients. By separating these into high-confidence and low-confidence groups, we identified patient-specific factors correlating with model confidence, revealing insights into temporal dynamics of gait recovery and individual variability in adaptations following ACL reconstruction.

4. Results

4.1. Predictive modeling evaluation

Our predictive modeling evaluation, illustrated in Table 1, compared the performance of five machine learning models (SVM, Naive Bayes, Random Forest, KNN, and Neural Network) across three frequency ranges (8 Hz, 16 Hz, and 32 Hz) for two classification tasks in both walking and jogging activities. Across all scenarios, the models demonstrated different levels of accuracy, with most achieving above 80% accuracy for left–right classification and above 90% for healthy-injured classification.

The KNN classifier consistently achieved the highest accuracy scores in all scenarios. For left–right classification, KNN reached a peak accuracy of approximately 93% during walking and 98% during jogging using the 16 Hz frequency range. For healthy-injured classification, KNN's performance was even more impressive, achieving approximately 98% accuracy for walking and 99% for jogging, both with the 16 Hz filtered data. While other models showed competitive performance in certain scenarios, their

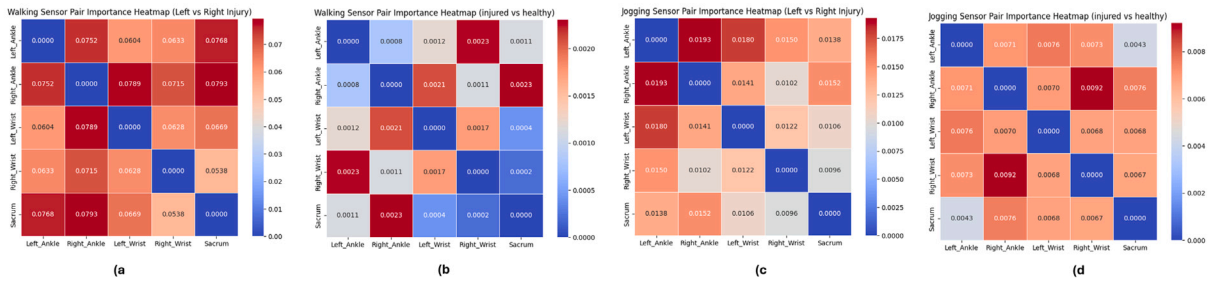

Fig. 2. Heatmap charts showing the relative importance of sensor pairs on predicting outcomes for (a) left vs right injury while walking, (b) left vs right injury while jogging, (c) healthy vs injured while walking, and (d) healthy vs injured while jogging.

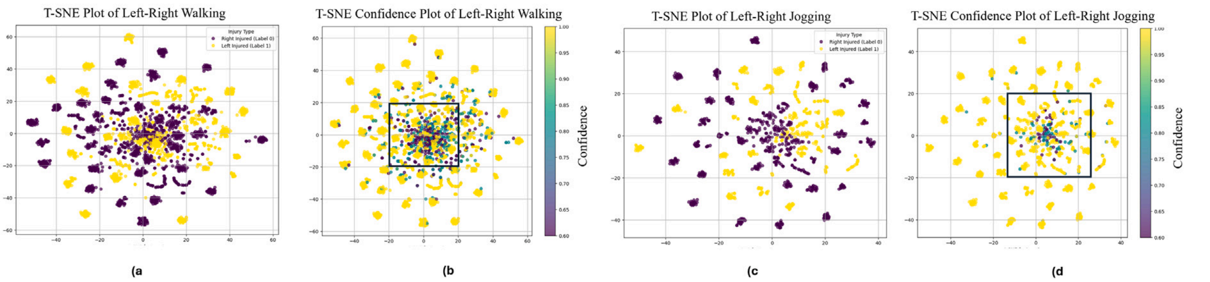

Fig. 3. T-sne dimension reduction plot using PSI feature matrix colored by (a) left or right injury (walking phase) (b) model confidence (walking phase), (c) left or right injury (jogging phase) (d) model confidence (jogging phase),.

accuracy was generally lower and more variable across different frequency ranges. The 16 Hz frequency range for PSI feature calculation typically yielded the highest accuracy scores across most models and tasks.

4.2. Task-specific sensor pair importance analysis

Fig. 2 presents heatmap charts illustrating the relative importance of sensor pairs in predicting outcomes during walking and jogging activities. The heatmaps were generated by accumulating permutation importance scores for each feature dimension, mapped back to the original sensor reading dimensions.

For the left vs. right injury classification during walking (Fig. 2a), five sensor pairs showed importance scores above 0.07, indicated by dark red colors. In contrast, the jogging condition (Fig. 2b) displayed only two sensor pairs with importance scores above 0.0175. This difference suggests that walking data exhibits more variability in discriminative patterns between left and right injuries compared to jogging.

In the healthy vs. injured classification, a similar trend was observed. The walking condition (Fig. 2c) showed two sensor pairs with importance scores above 0.002, while the jogging condition (Fig. 2d) had only one sensor pair exceeding the dark red threshold (> 0.008). This pattern aligns with the left vs. right injury analysis, indicating fewer highly important sensor pairs for jogging compared to walking. These observations correspond with the model prediction accuracies reported earlier, where jogging classifications generally achieved higher accuracy than walking classifications.

4.3. Dimension reduction examination

To visualize the high-dimensional PSI feature matrix, we employed t-SNE dimension reduction. In Fig. 3, each data point represents an instance from the PSI feature matrix, with points in the right column colored by the KNN model's output probability.

The t-SNE plots revealed that inner circles predominantly contain instances where the model exhibits low confidence in distinguishing between classes. Notably, jogging phase plots showed more distinctly separated clusters with reduced class overlap compared to walking, suggesting that jogging captures better global data structure. This improved cluster separation aligns with the superior model performance in jogging conditions relative to walking.

4.4. Patient-specific factor analysis

We examined patient-specific factors associated with data points in low-confidence regions and high-confidence regions of the t-SNE plots, focusing on recovery duration. Box plots in Fig. 4 illustrate recovery duration distributions for walking, further divided into low-confidence and high-confidence regions. For left leg injury, low-confidence regions had a median recovery of approximately

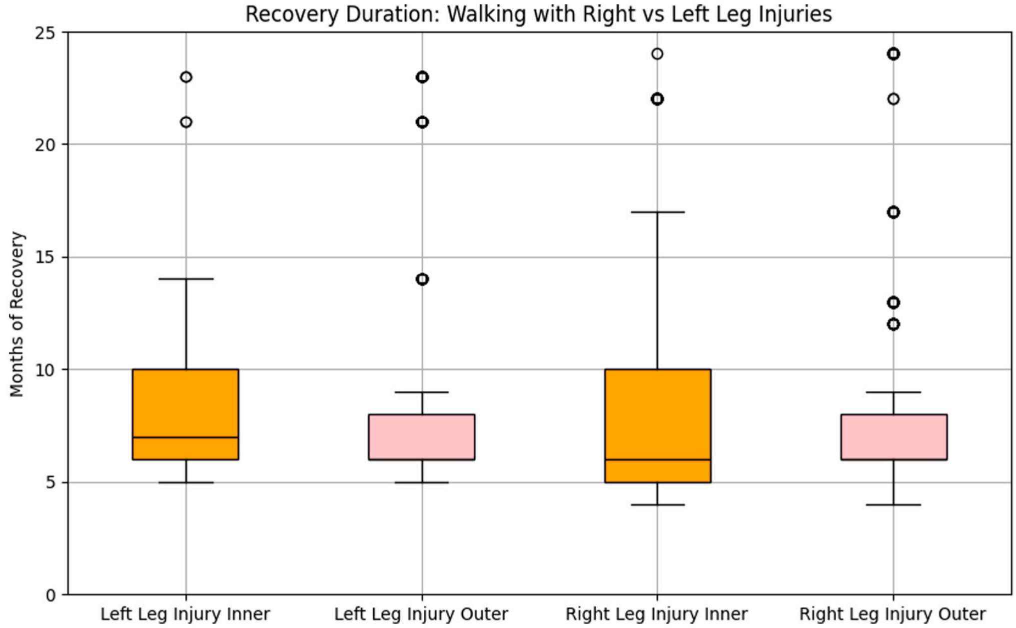

Fig. 4. Recovery duration for walking data, categorized by injured leg and location within or outside the low-confidence area (inner and outer regions).

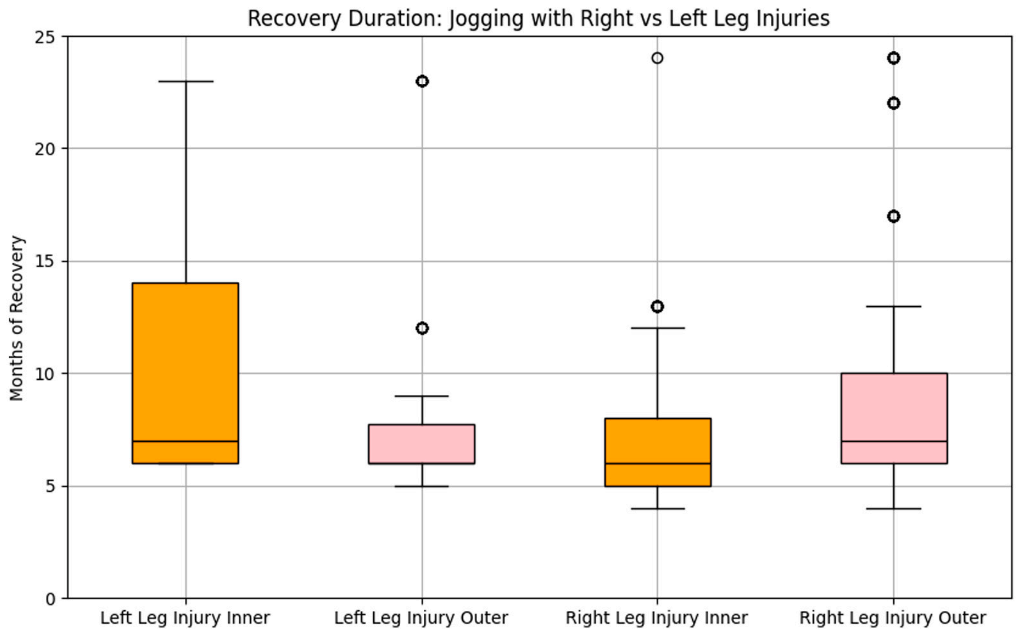

Fig. 5. Recovery duration for jogging data, categorized by injured leg and location within or outside the low-confidence area (inner and outer regions).

10 months (IQR: 8–13 months), while high-confidence regions showed a median of about 8 months (IQR: 6–10 months). Right leg injury during walking showed similar patterns. Fig. 5 displays recovery durations for jogging, with left leg injury following the same trend as walking. However, right leg injury in jogging exhibited a different pattern, with high-confidence regions showing a higher median recovery duration (approximately 10 months) compared to low-confidence regions (median of about 7 months), possibly due to outliers with recovery durations exceeding 20 months.

These findings suggest that participants with longer recovery durations generally exhibit gait patterns that are more difficult for the model to classify with high confidence. This trend supports the clinical understanding that as recovery progresses, movement patterns gradually normalize, becoming more similar to uninjured gait. The correlation between recovery duration and classification confidence provides quantitative evidence for rehabilitation progress, indicating that our model could potentially serve as an objective tool for tracking recovery timeline and informing return-to-sport decisions based on movement pattern normalization.

5. Discussion & conclusion

This study developed an explainable predictive model for ACL reconstruction classification through multi-modal analysis of gait dynamics and patient characteristics. Our findings supported our hypotheses: PSI-based features from IMU data effectively classified ACL reconstruction outcomes with 95.37% accuracy using KNN; sensor pair importance differed between walking and jogging tasks, with jogging showing more focused importance patterns; and recovery duration correlated with model confidence, suggesting gait patterns normalize over time post-reconstruction. The explainable nature of our approach offers significant clinical advantages by identifying key movement relationships affected by ACL reconstruction and providing quantitative tools to track patient progress.

Despite limitations including small sample size, reliance solely on IMU data, and single time point analysis, our novel approach demonstrates the potential of combining multi-modal gait analysis with explainable machine learning for ACL reconstruction assessment. By providing interpretable results that enhance clinical decision-making, this work establishes a foundation for future research integrating quantitative gait analysis into clinical practice, potentially enabling more personalized rehabilitation protocols and data-driven return-to-sport criteria.

CRediT authorship contribution statement

Xishi Zhu: Conceptualization, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft. **Ryan Henry:** Methodology, Visualization. **Emily Jackson:** Formal analysis, Methodology. **Joe M. Hart:** Conceptualization, Data curation, Methodology, Resources, Writing – review & editing. **Jiaqi Gong:** Conceptualization, Investigation, Methodology, Project administration, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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